2026

C Company AI Supply Chain Carbon Management System Strategic Implementation Plan Report

Group：我的未来我做组

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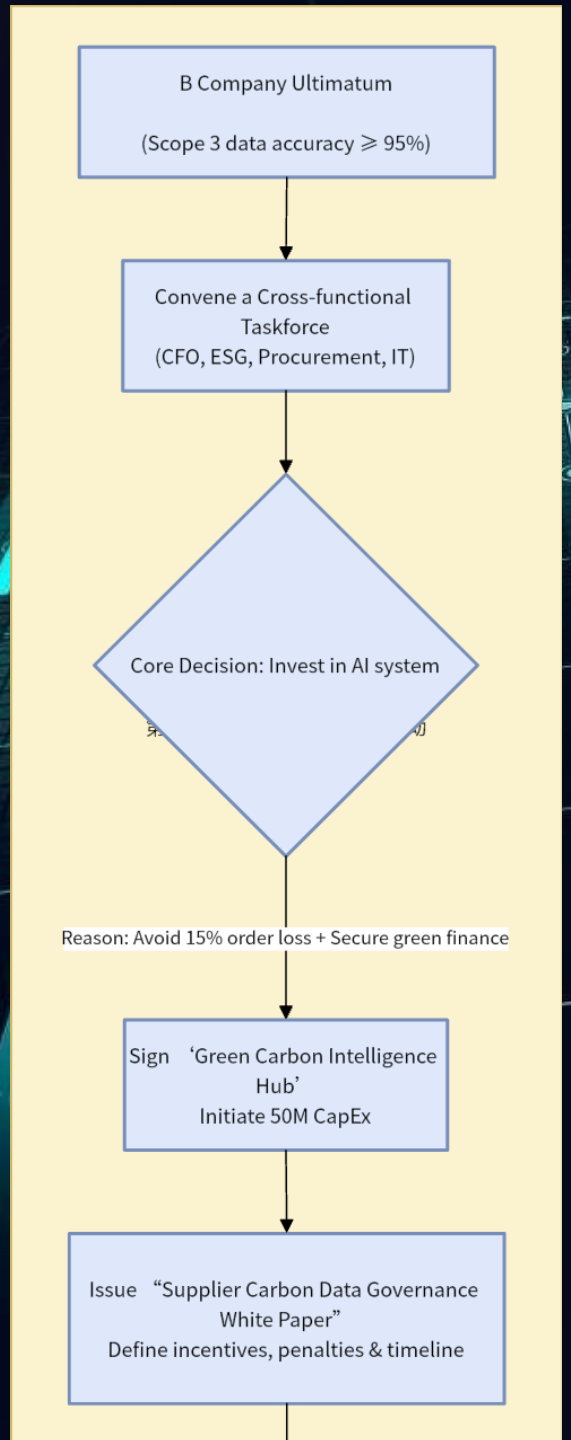
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Project Background and Strategic Necessity Analysis



External Pressure: Customer Compliance Ultimatum and Regulatory Trends

ISSB Standard Core Requirements

The International Sustainability Standards Board (ISSB) issued IFRS S1 (General Requirements) and IFRS S2 (Climate-related Disclosures) standards in 2023, mandating disclosure of Scope 1 and 2 emissions, with Scope 3 adopting the "comply or explain" principle and requiring third-party assurance. Non-compliant enterprises face investor pressure, increased financing costs, and customer order reduction risks.

EU CBAM Policy Impact

The EU Carbon Border Adjustment Mechanism (CBAM) began trial operation in October 2023 and will be officially implemented in 2026. Consumer electronic products exported by Company C to the EU require declaration of carbon emissions data; inaccurate data will result in carbon tariffs. CBAM requires Scope 3 data accuracy ≥ 90%, similar to Company B's requirements, creating compounding compliance pressure.

Regulatory Trends

Global ESG regulation is tightening. The U.S. SEC, EU, and Hong Kong have all issued mandatory disclosure requirements. Supply chain carbon emission data transparency will become a threshold for global manufacturing; non-compliant enterprises will be excluded from mainstream supply chains.

Requirement	Specific Content	Deadline	Consequence of Non-Compliance
Accuracy Requirement	Scope 3 supply chain carbon data traceability accuracy ≥ 95%	By end of 2026	15% order reduction in FY2027
Assurance Standard	Issue assurance report compliant with ISSB (International Sustainability Standards Board) standards	By end of 2026	Revenue loss of RMB 24.84 billion
Data Transparency	Full coverage of 2,500 suppliers	By end of 2026	Termination of strategic partnership
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Core Crisis

Data Quality Crisis: Third-party audit sampling indicates supplier data error and falsification rates as high as 38%, indicating systemic "supply chain greenwashing."

Management Blind Spot: Data collection from 2,500 suppliers is primitive: manual Excel filing at month-end, email aggregation, with no real-time monitoring.

Departmental Resistance: The Procurement Department opposes the mandatory AI system; 30 core suppliers threaten to cut off supply (potentially causing an 8% revenue decline, RMB 18.4 billion).

Technology Blind Spot: Suppliers may purchase diesel privately via personal WeChat for power generation, completely bypassing corporate accounts; traditional SaaS systems cannot capture such documents.

Emission Category	Emissions (10,000 tons CO_2e)	Percentage	Data Collection Status	Accuracy
Scope 1 (Direct Emissions)	15.0	5%	On-site direct measurement (diesel, natural gas)	99%
Scope 2 (Purchased Electricity)	45.0	15%	Calculated based on grid electricity bills and green energy transaction vouchers	98%
Scope 3 (Supply Chain Indirect)	240.0	80%	Manual Excel filing, third-party audit sampling	62%

Quantification of Order Loss Risk

Amplification Effect: A 15% order loss results in a 148.3% net profit decline, far exceeding the order loss rate.

Survival Crisis: A 15% order loss will cause RMB 5 billion in losses, directly piercing Company C's thin 4.5% net profit margin baseline.

Break-even Warning: A 10% order loss already brings Company C close to the break-even point (with only RMB 0.12 billion net profit remaining).

Scenario	Order Loss Rate	Revenue Loss	New Net Profit	Net Profit Change	Profit Decline Percentage	Profit/Loss Status
Baseline (2025)	0%	RMB 0 billion	RMB 10.35 billion	-	-	Profitable
Scenario A	10%	RMB 16.56 billion	RMB 0.12 billion	-RMB 10.23 billion	-98.8%	Near break-even
Scenario B	15%	RMB 24.84 billion	-RMB 5.00 billion	-RMB 15.35 billion	-148.3%	Severe loss
Scenario C	20%	RMB 33.12 billion	-RMB 10.12 billion	-RMB 20.47 billion	-197.8%	Near bankruptcy

02

Industry Best Practices Reference: Deconstruction of Foxconn Industrial Internet's Model

Benchmark Selection Rationale

Foxconn Industrial Internet (Fii)'s Carbon Management Experience:

Before designing Company C's solution, we studied the proven practices of Foxconn Industrial Internet (Fii), a global EMS leader. Fii's challenges mirror Company C's: serving top clients (like Apple) with strict carbon goals and managing a complex global supply chain. Crucially, Fii turned this challenge into an advantage. Its "AI-driven Supply Chain Carbon Management" is a National Benchmark Case, proving the model works.



Placeholder for Fii's AI Implementation Flowchart for Carbon Tracking

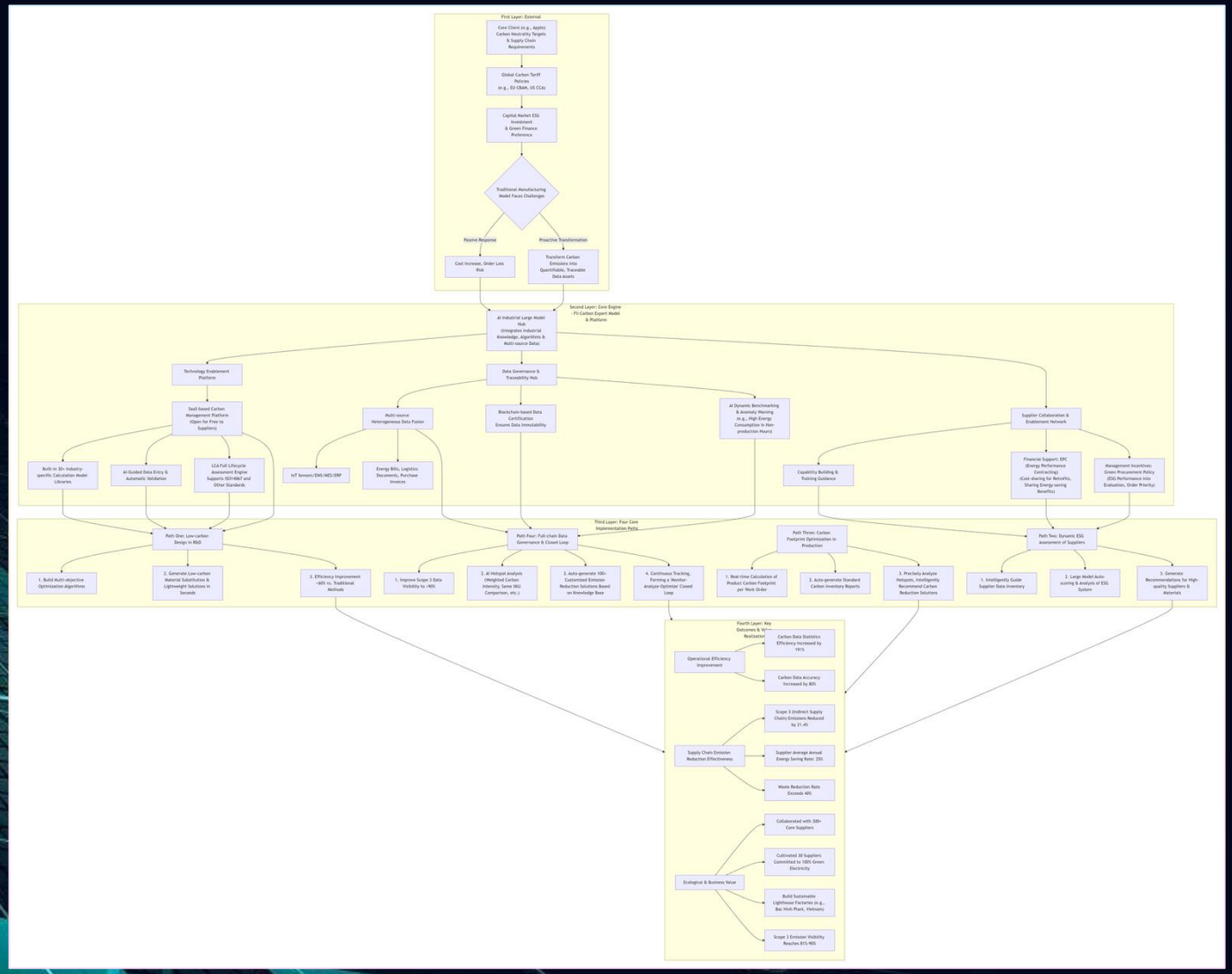
Implementation Pathways:

R&D End: Utilizes AI for low-carbon material substitution and lightweight design, improving efficiency by 60%.

Supplier End: Engages over 300 core suppliers through a free SaaS platform, training, and EMC (Energy Performance Contracting) financial support.

Production End: Achieves real-time calculation of carbon footprint at the work order level and automatic report generation.

Governance End: Increases Scope 3 data visibility to over 90% and utilizes AI for hotspot analysis and solution recommendation.



核心框架分析流程图 Core Framework Analysis:



Technology Enablement Platform:

Platform Architecture: Centered on the self-developed "Fii Carbon Expert" industrial large model, it builds on three pillars: Technology Enablement (SaaS Platform), Data Governance (Multi-source Data Fusion & Blockchain), and Supplier Collaboration (Training + Finance + Incentives).

Data Governance Engine

Supply Chain Collaboration

Building a Zero-Carbon Green Ecosystem Platform:

Industrial FII provides one-stop carbon neutrality services. We leverage AI-powered dual-carbon platforms to help enterprises achieve transparent carbon data, clear decarbonization paths, and supply chain collaboration, driving green growth.

At the 2025 Supplier Conference, we showcased 2024 results: 87 energy-saving and 16 waste-reduction projects delivered 25% energy savings, over 40% emission reduction, and more than RMB 20 million in economic benefits.

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Carbon Neutrality White Paper



推动价值链碳中和

工业富联携手供应商，通过准入机制、供应商管理及供应商激励等措施，提升全产业链的减排表现。

供应商准入管理

工业富联在供应商合规性和准入方面会进行严格考察，保证其符合环境规划与政府要求。公司对新供应商在准入阶段采集其气候变化管理的相关信息，包括能源使用、节能减碳、温室气体盘查与查证、节能减碳目标及方案、可再生能源的使用比例等。在进行供应商选择时，公司优先选择生产自动化程度高、能源使用效益较高的厂商。

供应商管理与合作

对于现存供应商，公司已制定一系列管理文件如《供应商碳管理要求》《供应商温室气体管理作业规范》等，并设置供应商碳管理系统（简称 GHG 系统）、供应商绩效管理等系统进行监控。GHG 系统于 2011 年正式上线，管理供应商碳数据，记录重点管控厂商的节能减排情况。

公司通过供应商 SER (Social and Environmental Responsibility，社会与环境责任）管理体系对供应商进行专项管理与监督。以企业社会责任、节能减排和绿色产品为三大管理主题，并下设多项细分议题对供应商进行管理与提升。

工业富联对其碳排放数据披露和环境相关认证进行严格管理，并加强与供应商沟通。对于高排放量的供应商，公司要求其依照 ISO14064-1《组织层面上对温室气体排放和清除的量化和报告的规范及指南》标准进行温室气体内部盘查并披露，并邀请第三方对数据进行查证。对部分高能耗供应商，公司要求其通过 ISO50001 能源管理体系认证。

2020 年，工业富联的关键供应商 100% 签署了《供应商社会及环境责任（SER）行为守则》。

Significant Dual Carbon Achievements

Technology Enablement Platform:

Carbon data statistics efficiency increased by 191%.

Supply chain (Scope 3) carbon emissions reduced by 21.4%.

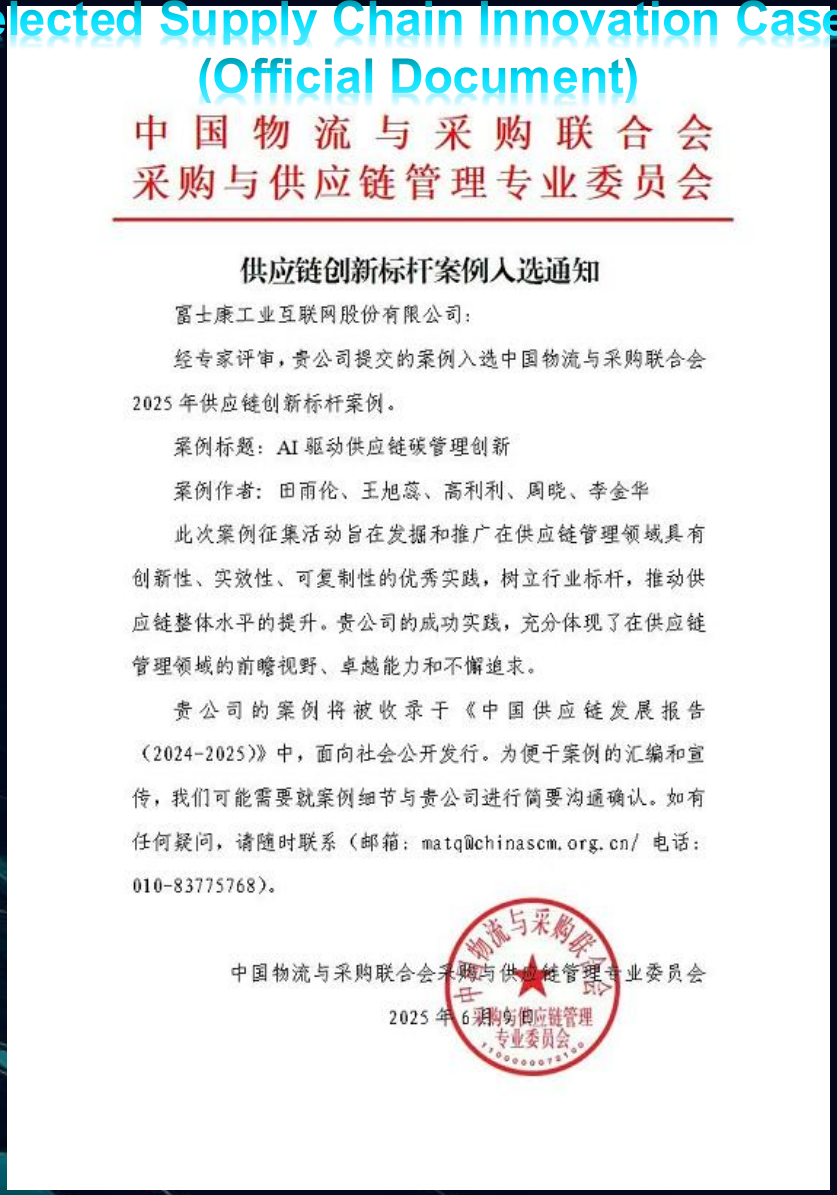
Supplier average energy-saving rate reached 25%.

Successfully cultivated 30 suppliers committed to using 100% green electricity.

Smart Carbon Management Increases Efficiency and Profits

Fii's practice proves that managing the carbon footprint of vast and complex supply chains through AI and platform-based methods is not only feasible but also leads to significant improvements in operational efficiency and commercial value.

Selected Supply Chain Innovation Cases (Official Document)



中国物流与采购联合会
采购与供应链管理专业委员会

供应链创新标杆案例入选通知

富士康工业互联网股份有限公司：

经专家评审，贵公司提交的案例入选中国物流与采购联合会2025年供应链创新标杆案例。

案例标题：AI驱动供应链碳管理创新

案例作者：田雨伦、王旭蕊、高利利、周晓、李金华

此次案例征集活动旨在发掘和推广在供应链管理领域具有创新性、实效性、可复制性的优秀实践，树立行业标杆，推动供应链整体水平的提升。贵公司的成功实践，充分体现了在供应链管理领域的前瞻视野、卓越能力和不懈追求。

贵公司的案例将被收录于《中国供应链发展报告（2024-2025）》中，面向社会公开发行。为便于案例的汇编和宣传，我们可能需要就案例细节与贵公司进行简要沟通确认。如有任何疑问，请随时联系（邮箱：matq@chinascm.org.cn/ 电话：010-83775768）。

中国物流与采购联合会采购与供应链管理专业委员会

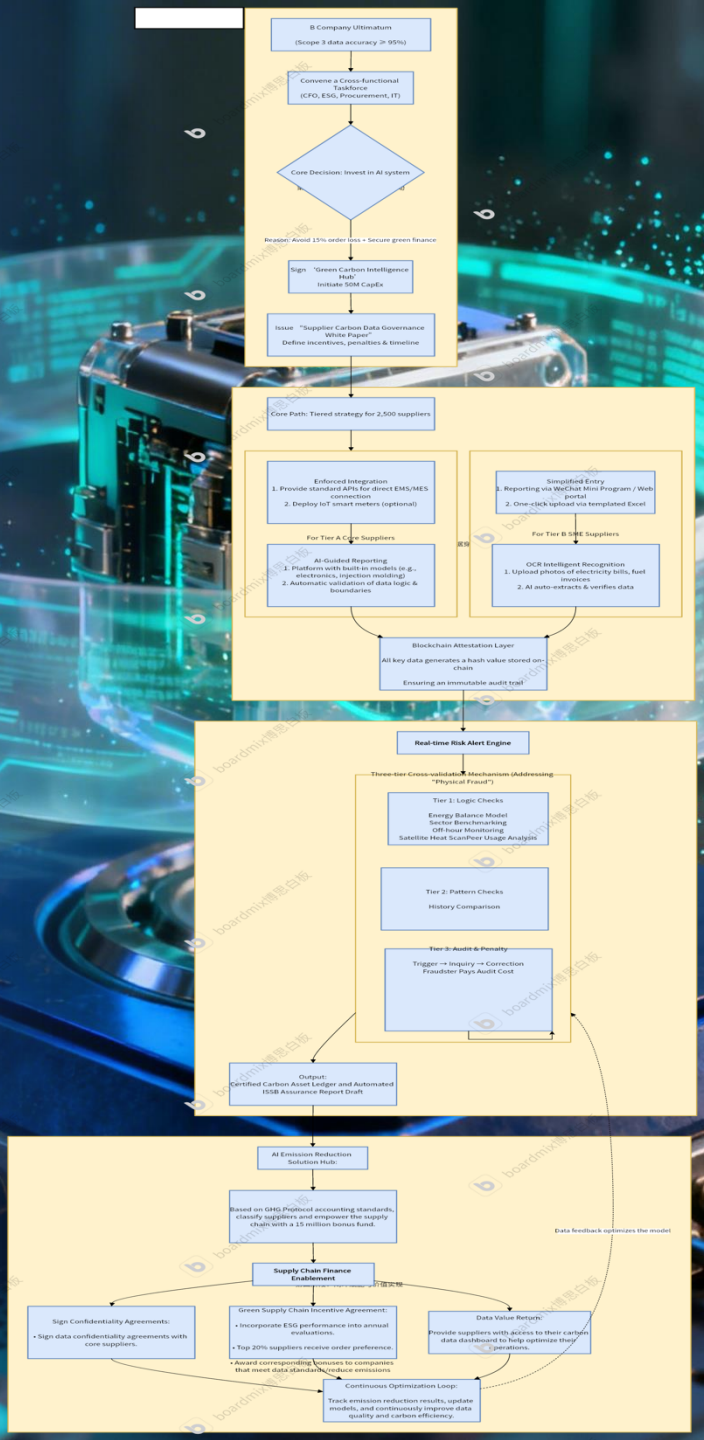
2025年6月9日


03
Company C Solution Blueprint—
"Green Carbon Intelligence

Designing the Four-Layer Implementation Blueprint

Overall Goal and Top-Level Design Principles

Building on Foxconn's proven model and tailored to Company C's critical "survival line" situation, we designed a four-layer blueprint. Its core is a closed-loop system featuring automated data flow, AI-driven verification, and collaborative benefit-sharing.



The "Green Carbon Intelligence Hub" Blueprint

Top Layer: Strategy
Goal: Secure orders at all costs.

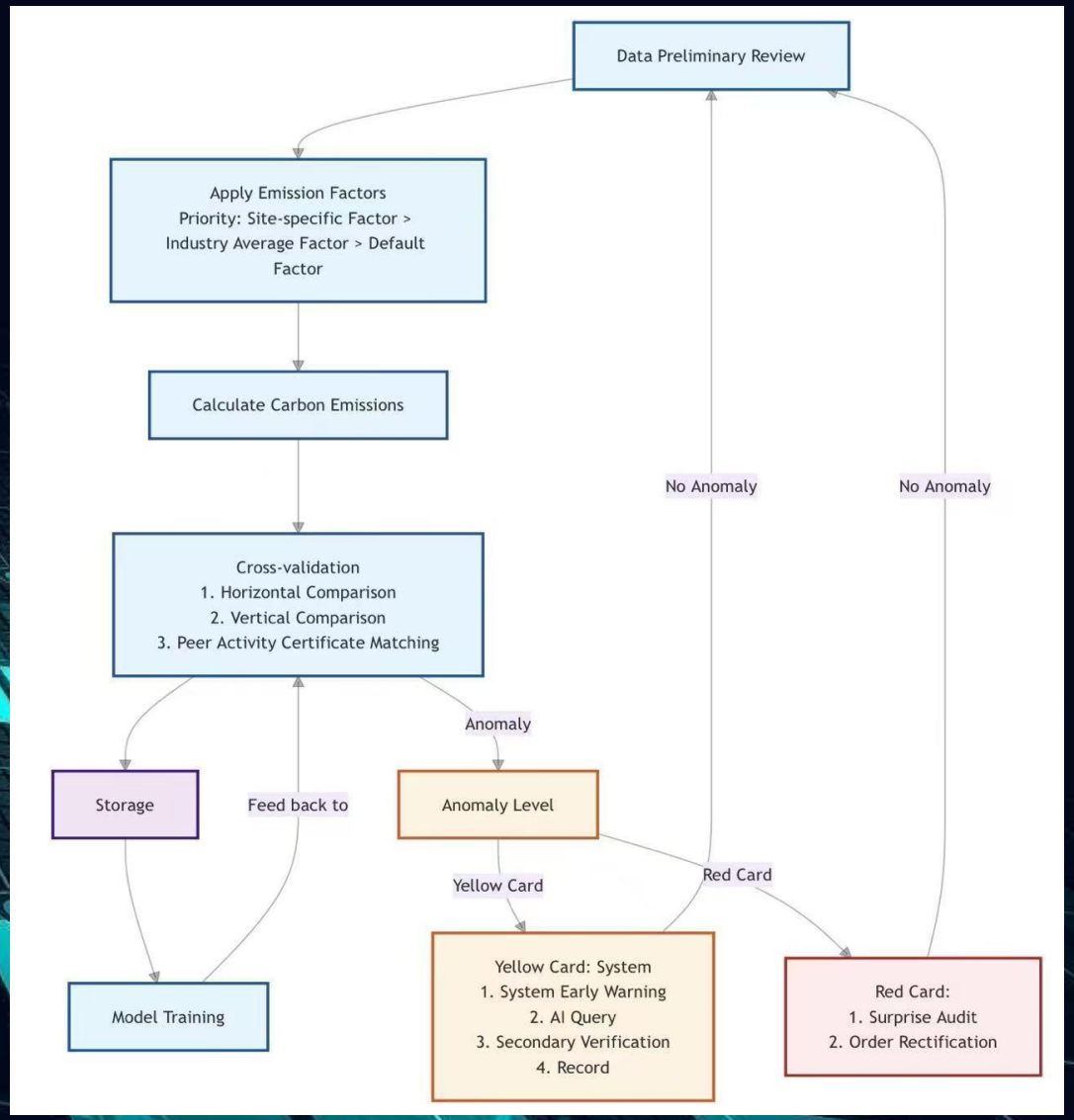
Core Layer: AI Hub
An engine that combines technology, data, and a collaborative network.



Action Layer: 4 Key Paths
Focuses on overcoming "physical fraud" through tiered supplier management and cross-checking data

Outcome Layer: Triple Value
Secures orders, unlocks green financing, and builds long-term data assets.

Flow chart


Data Preliminary Review
Apply Emission Factors
Priority: Site-specific Factor > Industry Average Factor > Default Factor
Calculate Carbon Emissions
Cross-validation
1. Horizontal Comparison
2. Vertical Comparison
3. Peer Activity Certificate Matching
No Anomaly
No Anomaly
Anomaly
Storage
Feed back to
Anomaly Level
Yellow Card
Red Card
Model Training
Yellow Card: System
1. System Early Warning
2. AI Query
3. Secondary Verification
4. Record
Red Card:
1. Surprise Audit
2. Order Rectification

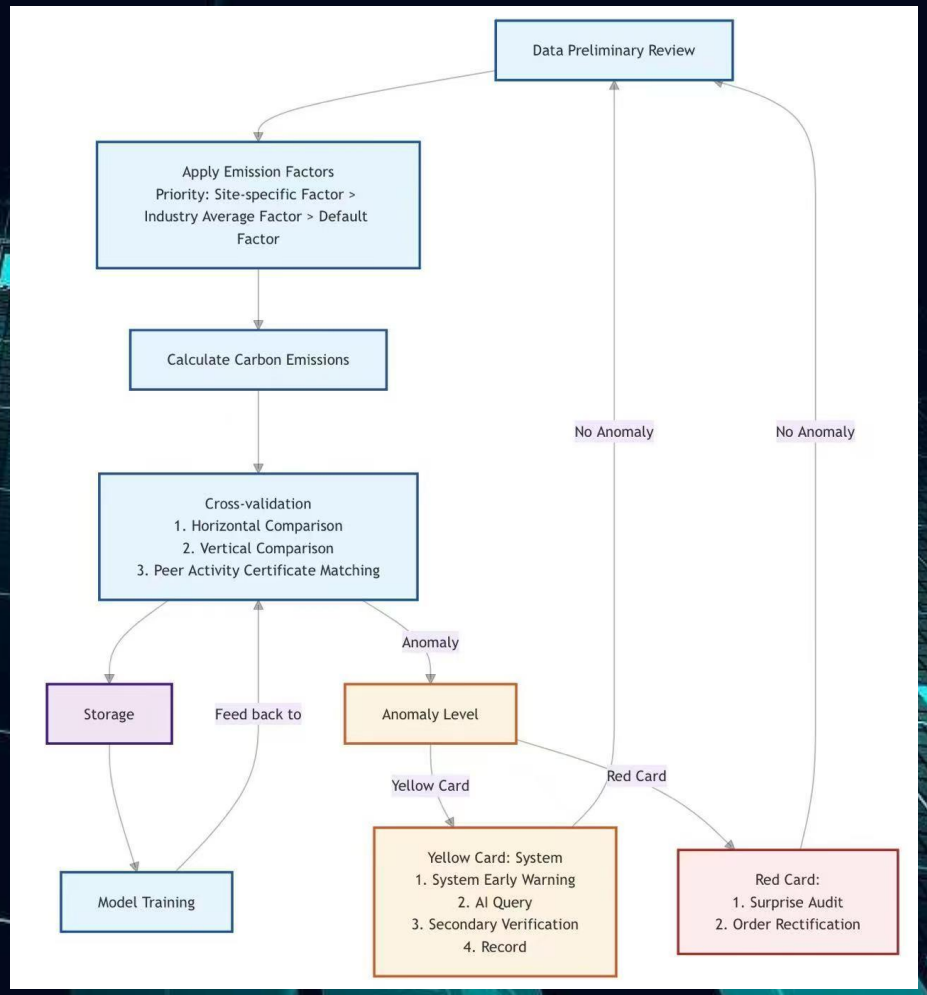
Supplier Classification and Management Strategy

Category A Supplier (Core Suppliers):Such as metal structural parts factories, PCB board factories, and chip agents. For such suppliers, it is Mandatory actual measurement. Requires direct system-to-platform integration of their production and energy management systems for real-time data transmission.

Category B (Suppliers General Suppliers) :Such as packaging material factories and cable factories. It adopts the hybrid mode of "AI-assisted filling + database calibration", and plans to help it establish actual measurement capabilities through training and empowerment

Category C Suppliers (Minor Suppliers) :Such as office supply suppliers and small logistics fleets. Low emissions and high management costs. The efficiency priority "database factor" method is used to directly call the average emission factor of the authoritative database for calculation, and the total trend change is monitored

The whole process control of data collection and verification



1. Initial verification during data extraction:

(1)The document itself is standardized

(2) Internal self-consistency

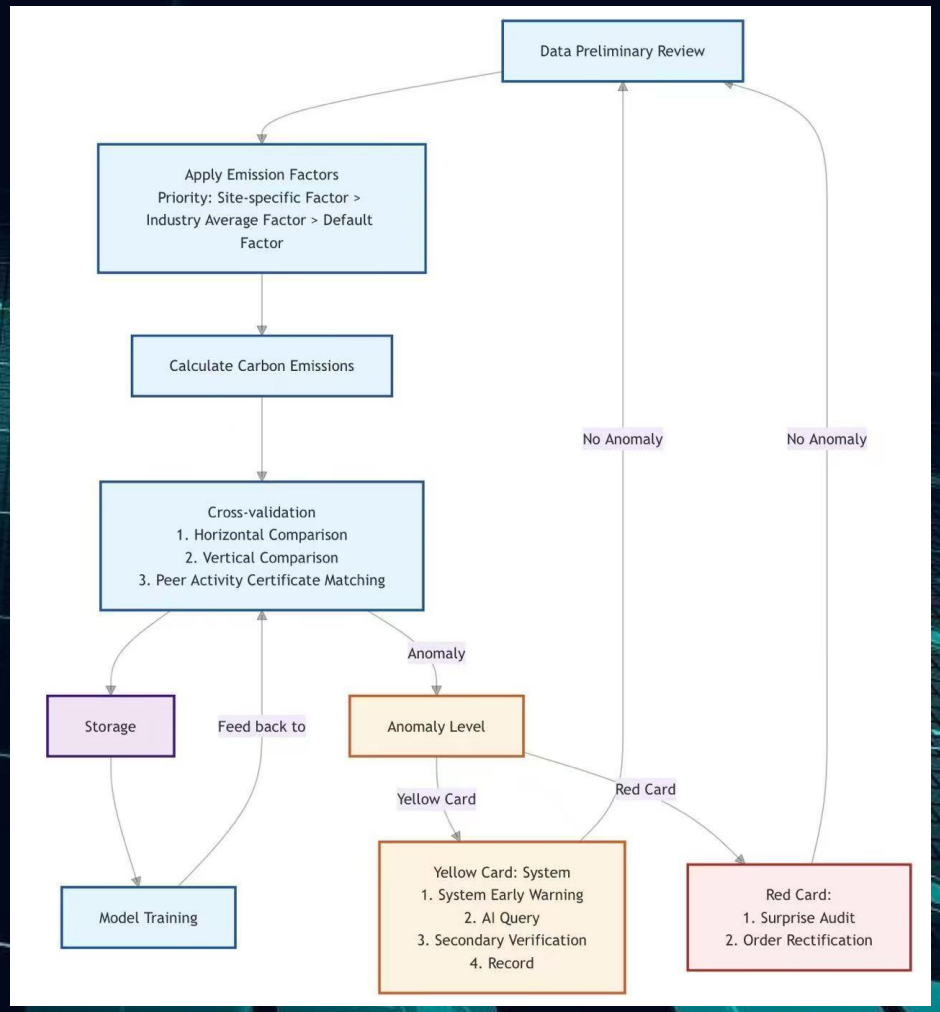
2. Define data boundaries: Classify the data into 15 categories of Scope 3 carbon emissions (classification criteria from the official website of GHG Protocol)

3. Priority of emission factor sources:

Location-specific factors> industry average factors> default factors

Industry average factor source: (China data source: CPCD, CLCD International data source: Ecoinvent, IPCC)

The whole process control of data collection and verification



1.Calculate carbon emissions by classification after data extraction:

Emissions = activity data × emission factors to calculate carbon emissions

2. Cross-verification

(1) Time-series consistency check: the same data source across time dimensions of consistency verification

(2) Industry benchmark comparison : the carbon emissions of the process are compared with the benchmark value of the same process in the same industry

(3) Multiple evidence matching for the same behavior

The whole process control of data collection and verification

(3) Multiple evidence matching for the same behavior

High-risk categories（1, 3, 4, 9）:

1:Purchased Goods and Services： inputs vs outputs

3.Fuel- and Energy-Related Activities: inputs vs outputs

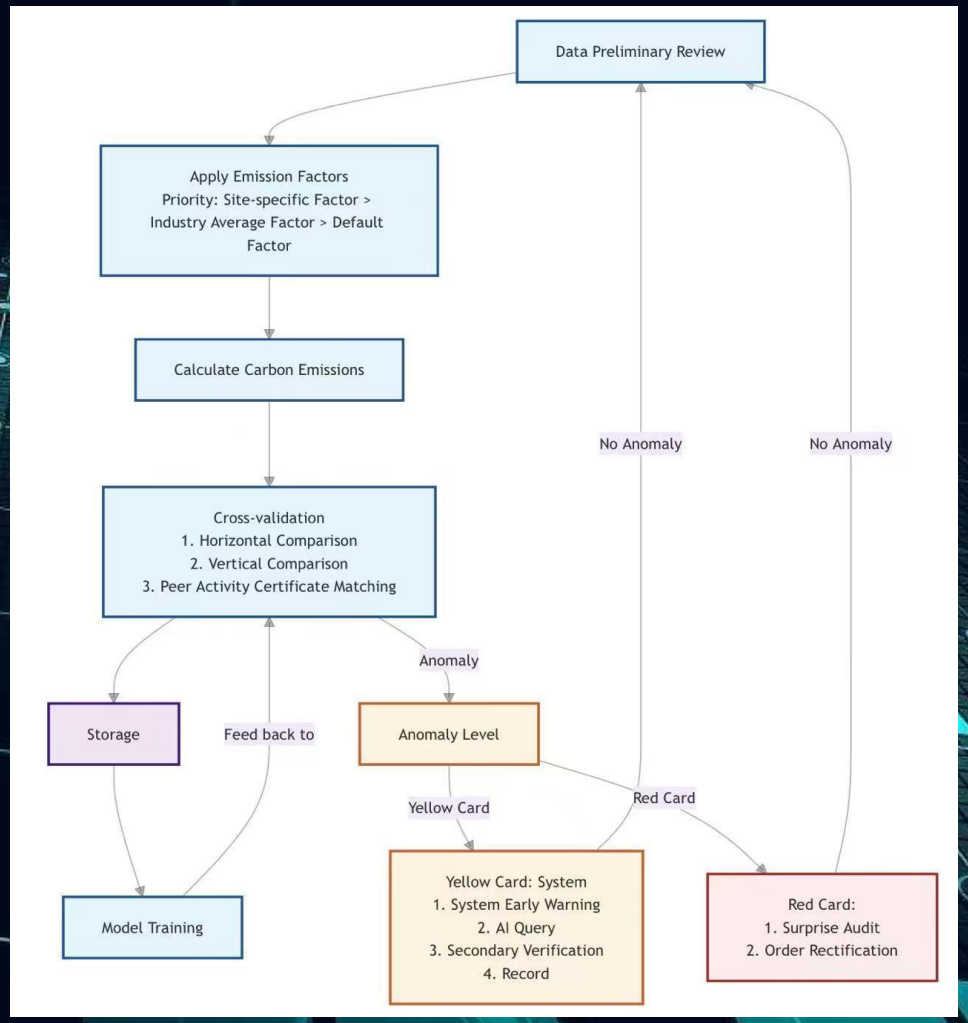
4.Upstream Transportation and Distribution： distance vs fuel consumption

9. Downstream Transportation and Distribution ： distance vs fuel consumption

Medium risk categories (2, 5, 6, 8, 13):business orders, financial invoices, and third-party records

For low-risk categories (7,10,11,12,14,15):Sampling verification， Macro trends

The whole process control of data collection and verification



1.Data hierarchical storage architecture

2. Establish a supplier's exclusive data warehouse Category:

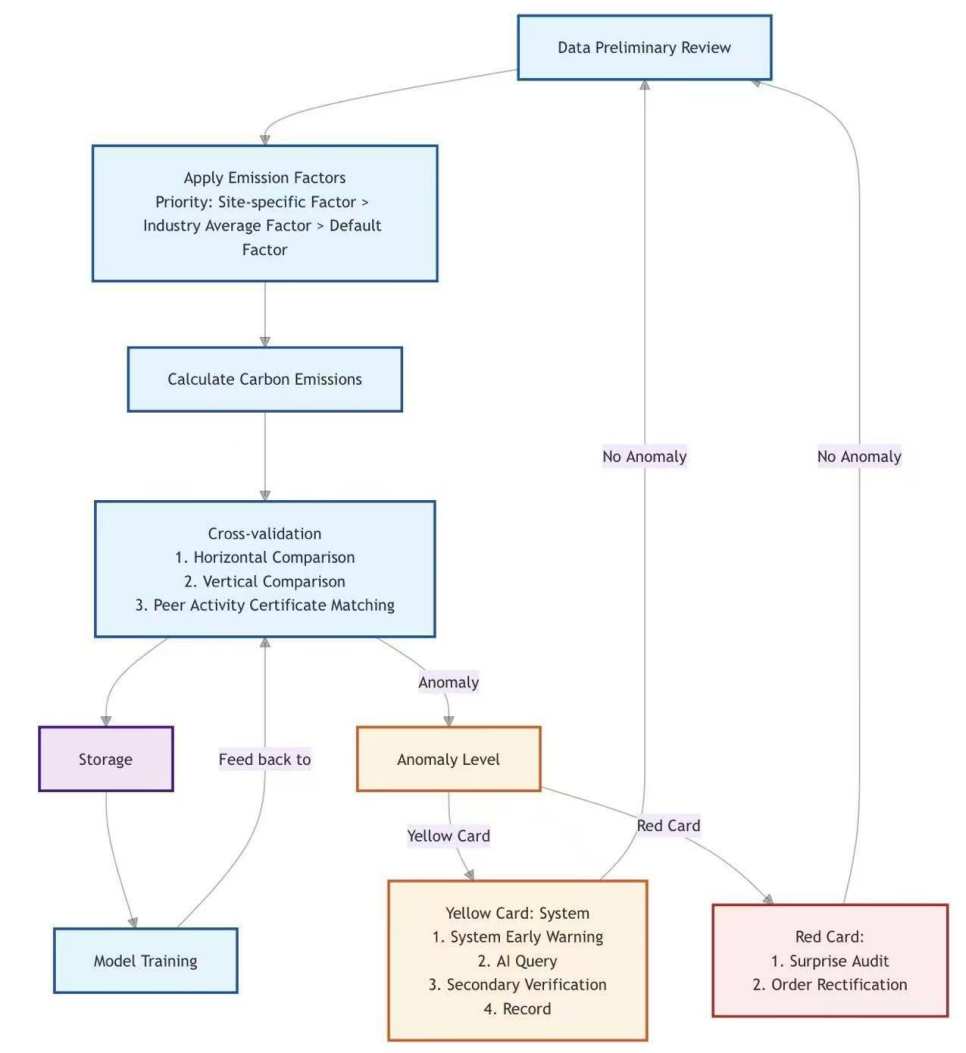
3. Model training Training objectives:

Customize a dual-objective model for a single supplier

(1)Data verification model: identify the anomalies of the supplier's carbon data (such as data falsification/reporting errors in specific emission categories)

(2) Carbon emission prediction model: Predict future carbon emissions based on its production/energy consumption data and identify emission reduction opportunities in advance.

Response measures to abnormal situations



Abnormal levels are divided into two levels: yellow cards and red cards

Yellow card (slight abnormality):

1. Automatic early warning of the system 2. AI intelligent questioning

3. Secondary cross-verification of the system

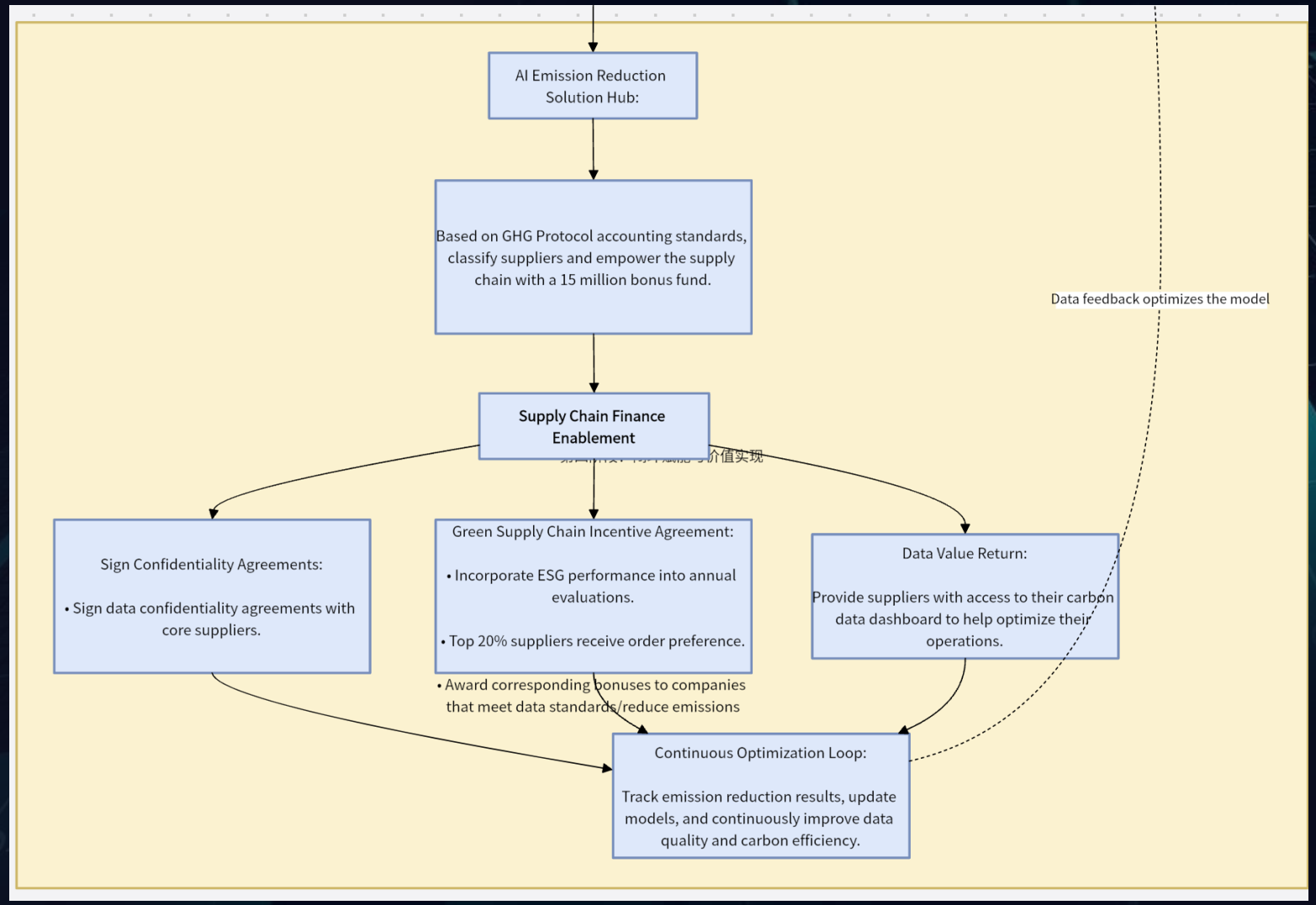
4. The situation is recorded in the database Red Card (Serious Abnormal):

1.Surprise audit

2. Order it to rectify

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Implementation Roadmap and Resource Assurance



Core Phases and Milestones

8-Month Countdown Action Plan

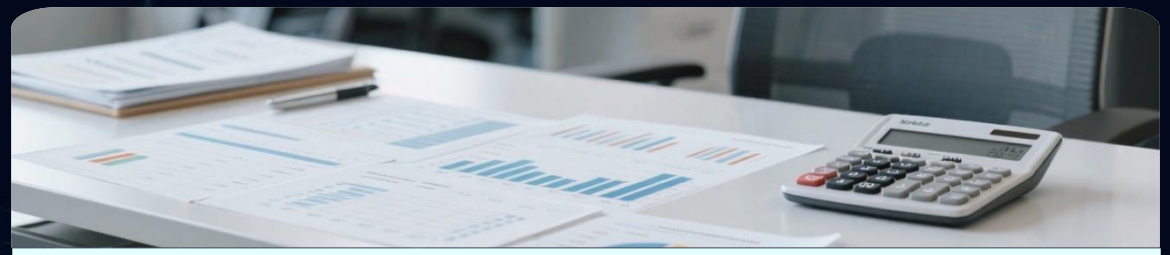
Phase No.	Phase Name	Timeline	Core Actions
1	Launch & Pilot	Months 1-2	Form a task force led by the CFO, connect our systems, and start with 50 cooperative suppliers.
2	Build & Break Ice	Months 3-4	Hold a big supplier meeting. Sign the first EMC deals with the toughest 3-5 suppliers, using real funding to break the deadlock.
3	Full Push	Months 5-6	Make the top 300 suppliers connect. Let the rest use simple reporting tools. Start full AI cross-checking of all data.
4	Check & Audit	Months 7-8	Bring in a third-party auditor early. Fix any data issues. Create a draft report that meets global standards.
5	Final Delivery	End of Month 8	Submit the report to Company B. At the same time, apply for

Cost estimate

Data comes from Exhibit4:

Program	Amount (¥)	Explanation
Capital expense	50,000,000	System setup and API development, amortised over five years at 10 million yuan per year
Operational expense	15,000,000	Costs of SaaS subscriptions and model training
Income (interest savings)	15,000,000	5 billion yuan in green loans, with interest rates reduced by 30 basis points, resulting in annual interest savings of 15 million yuan

ROI calculation



Annual net income：

Interest savings	¥15,000,000
less： Operating expenses	¥15,000,000
Depreciation	¥ 000,0 0
	- ¥10, 000

if we consider only the investment and annual returns (ignoring the impact of depreciation on cash flow):

ROI = annual returns / investment = 1,500 / 5,000 = 30%

meaning that 30% of the investment is recouped each year, and the payback period is approximately 3.33 years.



05 Negotiation Strategies and Key Contractual Points



Negotiation Strategy

——Specific Mechanisms to Alleviate Fear of"Exposed Costs Leading to Price Pressure"



- Institutional Safeguard
- Technical Safeguard
- Interest Hedging

Institutional Safeguard:

Signing"Data Use Limitation Agreement"

- Core Promise: Energy and material data collected by AI will only be used for carbon accounting and compliance reporting.
- Prohibited Uses: Such data cannot be used for price negotiation, cost analysis, or order adjustment.
- Recourse: If violated, suppliers may apply for arbitration or appeal to Company C’s board of directors.

Technical Safeguard:

Third-Party Independent Audit

- Independent Verification: Bring in an authoritative third party (e.g., Guangzhou Carbon Exchange) to verify data.

- Tiered Access:

ESG department + auditors: Can view full original data.

Procurement department: Can only see final rating results, not the original data.

- Effect: Technically prevents the procurement department from using carbon data to pressure prices.

Interest Hedging:

Replace"Price Pressure"with"Priority Procurement"

- A-Level Supplier Privilege: Under the same quotation, Company C gives priority to placing orders with them.

- Commitment: Company C will not lower purchase prices just because a supplier joins the system. Instead, it increases supplier revenue through order preference.

- Logic Shift: Turn the fear of "exposed costs" into an opportunity for "green advantage."

contract

Green Supply Chain Incentive Agreement

Article 1： Purpose of Cooperation
Article 2： Data Access Requirements
Article 3： Supplier Scoring Mechanism
Article 4： Incentive Measures
Article 5： Supplier Obligations
Article 6： Audit and Verification
Article 7： Adjustment and Termination of Incentives
Article 8： Term and Termination of the Agreement
Article 9: Dispute Resolution
Article 10: Miscellaneous

Data Confidentiality and Use Restriction Agreement

Article 1： Definition of Confidential Information
Article 2： Restrictions on Data Use
Article 3： Confidentiality Obligations and Tiered Access
Article 4： Liability for Breach
Article 5： Term and Termination of the Agreement
Article 6： Dispute Resolution
Article 7： Miscellaneous

Design of a Supplier Rating System

Two-dimensional scorecard

In accordance with the requirements for ‘management systems’ and ‘supplier responsibility’ set out in the Foxconn Industrial Internet guidelines, and in line with the objectives for carbon data transparency, a two-dimensional scorecard has been designed comprising two dimensions: ‘data transparency’ and ‘carbon performance’. The criteria for the carbon performance score are based on international ESG rating standards, with a maximum score of 100 points.

Design of a Supplier Rating System

Details of Two-dimensional scorecard

Two-dimensional scorecard				
Level 1 dimension	**Percentage**	**Level 2 dimension**	**Marks**	**Marking criteria**
Data transperancy	60%	Data integration integrity	20	Have all underlying source documents—such as electricity bills, purchase invoices and logistics documents—been integrated into the AI system on schedule?
		Data accuracy	20	The data has been cross-checked (e.g. against yield-to-energy consumption ratios) and no anomalies were found; it has not been flagged by the AI system as 'suspected of being falsified'.
		Exceptional response times	10	Response times and correction lead times for AI system alerts
		Backfilling of historical data	10	Have the carbon data working papers for the past 12 months been completed?
Carbon performance	40%	Environment (E)	28	Details in the next Sheet
		Society (S)	6	Details in the next Sheet
		Govermance (G)	6	Details in the next Sheet

Design of a Supplier Rating System

Details of carbon performance scorecard

carbon performance scorecard			
Level 1 dimension	Level 2 dimension	Marks	Relate to ESG
Environment（E)	Year-on-year decrease in carbon emissions per unit	12	CDP Climate Change Questionnaire, Science-Based Targets Initiative (SBTi)
	Proportion of green electricity/renewable energy used	6	RE100 Initiative, TCFD Recommendations
	Energy Management System Certification	4	MSCI Environmental Pillar
	Waste Management and the Circular Economy	4	Circular economy indicators
	Water resource management (where applicable)	2	CDP Water Security Questionnaire
Society（S)	Employee Carbon Awareness Training	4	Employee engagement metrics
	Social Responsibility in the Supply Chain	2	Supply Chain ESG Management
Governance（G)	Internal Auditing and Disclosure of Carbon Data	4	Transparency and accountability
	Identifying and Addressing Climate Risks	2	Risk Management

Design of Incentive and Penalty Mechanisms

The annual interest savings of 15 million yuan will be allocated **across three main funds** to create a closed-loop system of **positive incentives**.

Design of Incentive and Penalty Mechanisms

Fund Pool 1: Supply Chain Finance Interest Subsidy Fund (8 million yuan per year)
Purpose: To provide loan interest rate subsidies to suppliers connected to the system and rated A or B
Operational Approach: Company C, in collaboration with partner banks, links supplier financing rates to carbon ratings

Fund Pool 2: Carbon Emissions Reduction Incentive Fund (5 million yuan annually)
Purpose: To provide cash rewards to the top 10 suppliers based on annual carbon performance rankings
Rationale: Similar “green supply chain bonus pools” have already emerged in the automotive and electronics sectors, such as Apple’s Clean Energy Fund.

Fund Pool 3: Capacity Building and Empowerment Fund (2 million yuan per year)
Purpose:① To provide small and medium-sized suppliers with free carbon accounting training and carbon management software tools (excluding core data)②Organise ‘green electricity group purchasing’: Company C will bring together 30 core suppliers to collectively purchase green electricity from an electricity supplier, thereby securing volume discounts (referencing the AstraZeneca case, where prices were reduced by 80%).

2026.4.2

Thanks for your listening

